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Cloud Security with AWS IAM

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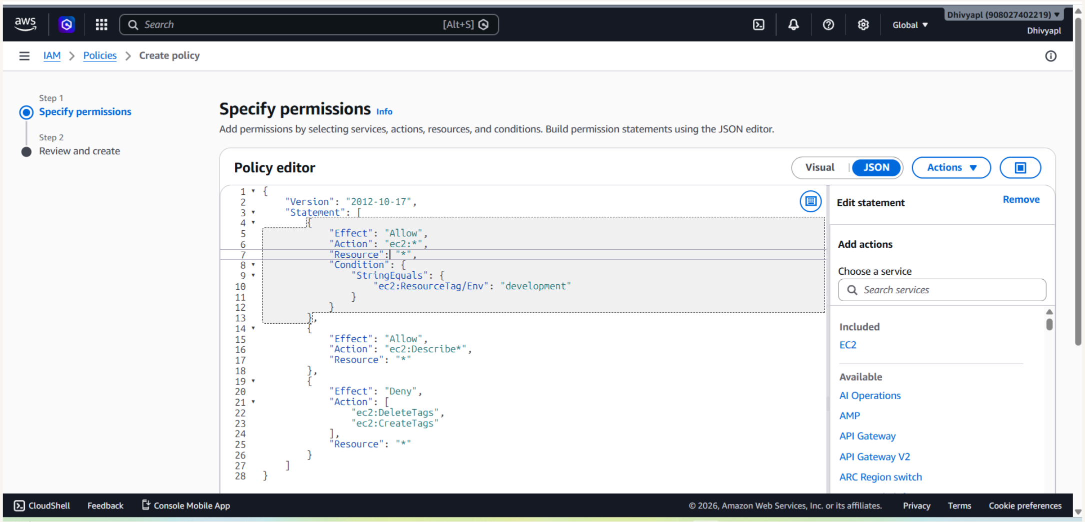


Introducing Today's Project!

Project overview

In this project, I will demonstrate to create a IAM Policy, IAM User and groups and I'm doing this project to learn AWS

Tools and concepts

Services I used were EC2, IAM Users, Groups and policy Key concepts I learnt include how to create start stop ec2 instance and how to create policy and how to create users and gropus and how to tag ec2 instaces and Account alias

Project reflection

This project took me approximately 2 hous The most challenging part was applying permissions to the users. It was most rewarding to see when the iam policy has been created

Tags

What I did in this step

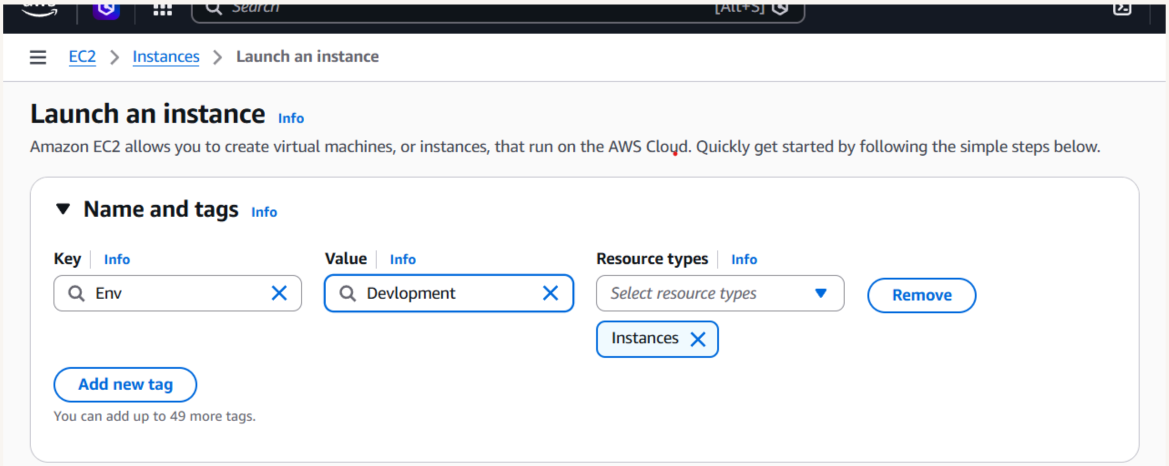
In this step, I will lanuch anEC2 Instance because to increase the computing power.

Understanding tags

Tags are like lables used in the aws resources and when we need to apply policies and cost optimization it will easy filter and apply accordingly

My tag configuration

The tag I’ve used on my EC2 instances is called ENV. The value I’ve assigned for my instances are Production and development


EC2 > Instances > Launch an instance
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Key Info
Env
Value Info
Devlopment
Resource types Info
Select resource types
Remove
Instances
Add new tag
You can add up to 49 more tags.

IAM Policies

What I did in this step

In this step, I will create an IAM policy because to restrict access to our intern from using the production environment and giving access to the Development environment

Understanding IAM policies

IAM Policies are the rules to set the users , groups and services to which resources they have the access

The policy I set up

For this project, I’ve set up a policy using JSON

Policy effect

I’ve created a policy that allows users to start or stop the ec2 servives those are tagged under Env-Development

Understanding Effect, Action, and Resource

The Effect, Action, and Resource attributes of a JSON policy means policy allowing the acctions

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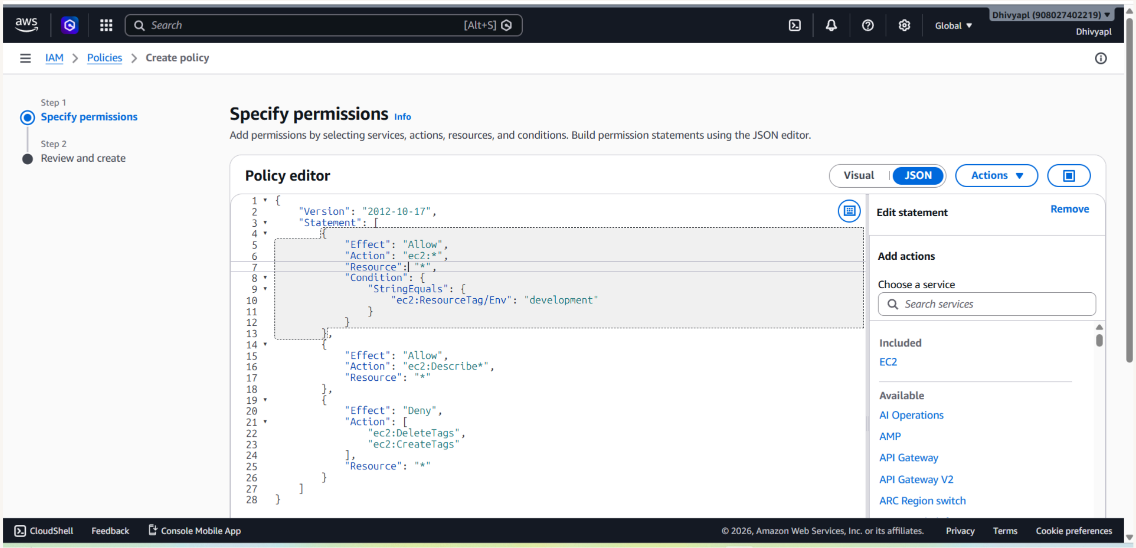
My JSON Policy



Account Alias

What I did in this step

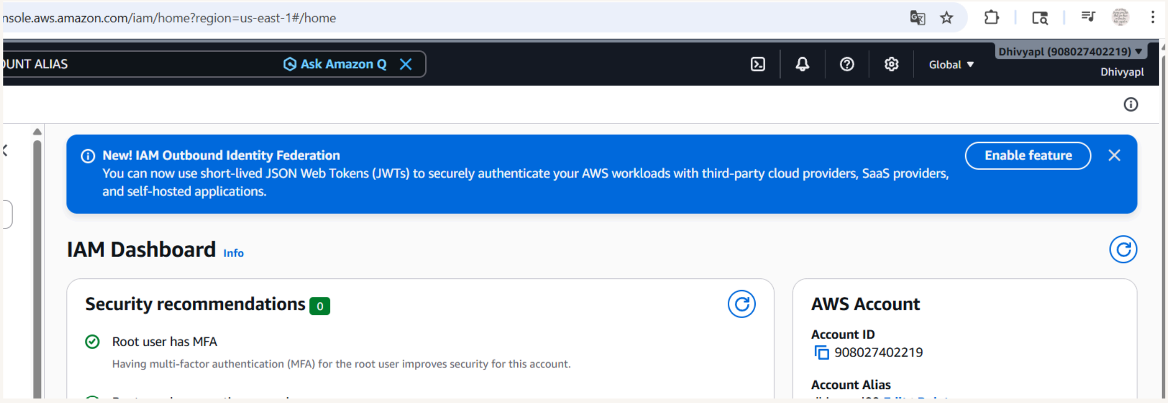
In this step, I will create account alias because to give an uniqe name instead of giving account ID

Understanding account aliases

An account alias is a freindle name for ur account which displays in the url when u are sharing ur account

Setting up my account alias

Creating an account alias took me less than minute Now, my new AWS console sign-in URL is https://dhivya-pl99.signin.aws.amazon.com/console


nsole.aws.amazon.com/iam/home?region=us-east-1#/home
Ask Amazon Q
Global
Dhivyapl (908027402219)
Dhivyapl
New! IAM Outbound Identity Federation
You can now use short-lived JSON Web Tokens (JWTs) to securely authenticate your AWS workloads with third-party cloud providers, SaaS providers, and self-hosted applications.
Enable feature
IAM Dashboard
Info
Security recommendations
0
Root user has MFA
Having multi-factor authentication (MFA) for the root user improves security for this account.
AWS Account
Account ID
908027402219
Account Alias

IAM Users and User Groups

What I did in this step

In this step, I will create users and groups because my intern should not have access to my own accont... So I need to create username and paswords for them

Understanding user groups

IAM user groups are collections / folder of users which is used to give permissions as a set.

Attaching policies to user groups

I attached the policy I created to this user group, which means the interns are capable to access my ec2 dev instance only

Understanding IAM users

IAM users are the persons who can access your aws acount services

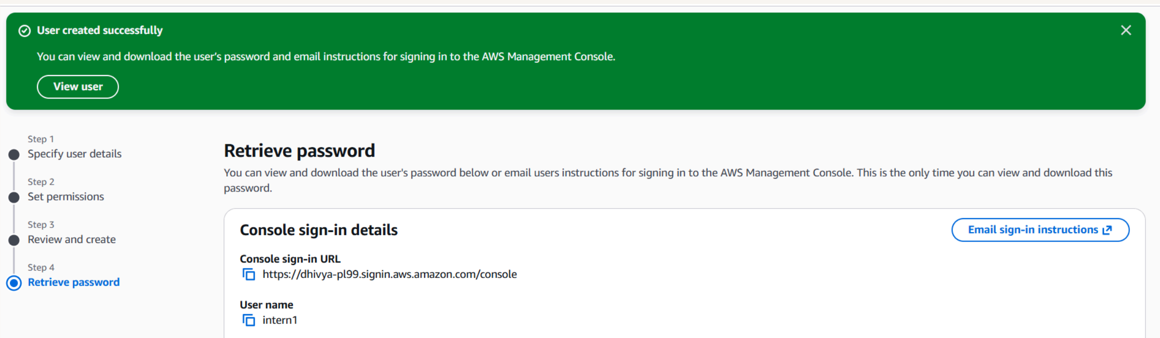
Logging in as an IAM User

Sharing sign-in details

The first way is .csv (Contaisn Ussers sign in url, Username, Password) and sending a mail with sign in details

Observations from the IAM user dashboard

Once I logged in as my IAM user, I noticed that most of the services are not allowed to view or edit.This was because i hav give access to only ENV-DEV EC2 Instance alone



Testing IAM Policies

What I did in this step

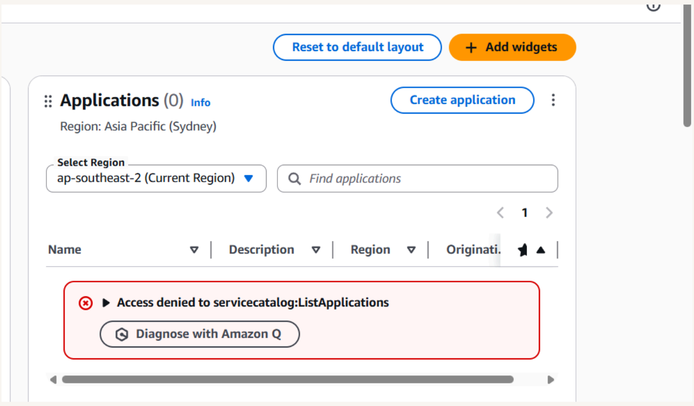
In this step, I will try to sign in with my IAM User intern1 cred because to check with they do not have the access to the Prod env ec2 instance

Testing policy actions

I tested my JSON IAM policy by stopping the ec2 instances

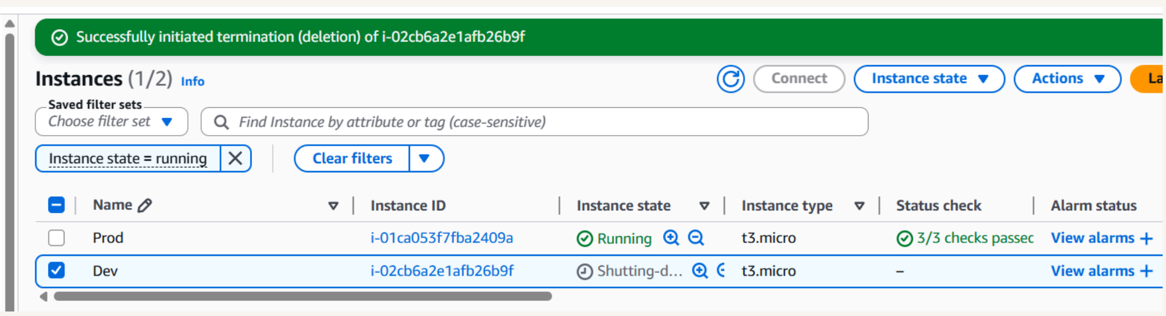
Stopping the production instance

When I tried to stop the production instance it throwed an error and This was because the users are not having access to prod env



Stopping the development instance

Next, when I tried to stop the development instance it stopped and This was because the users has the access to start or stop the env - dev tagged ec2 instance



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